Unit 3: Exponential and Logarithmic Functions

Honors Precalculus

Student

Contents of this unit:

Contents

3.1 Exponential Functions and Their Graphs

Objectives

- Recognize and evaluate exponential functions with base b and base e.
- Graph exponential functions and identify key features: domain, range, intercepts, and asymptotes.
- Distinguish between exponential growth and exponential decay.
- Analyze transformations of exponential functions.
- Write exponential equations given key points or conditions.
- Use compound interest formulas to solve financial problems.

Exponential Functions

An **exponential function** with base ________ is a function of the form

$$f(x) = b^x$$

where $b > 0$, $b \neq 1$, and x is any real number.

Unlike polynomial functions where the base is a variable and the exponent is a constant (e.g., x^2), an exponential function has a ________ base and a ________ exponent.

Exponential Growth and Decay

Let $f(x) = b^x$ be an exponential parent function:

- **Exponential Growth:** If ________, the function grows as x increases. The quantity grows when its rate of increase is proportional to its current value.
- **Exponential Decay:** If ________, the function decays as x increases. The quantity decays when its rate of decrease is proportional to its current value.

Fact

Parent Exponential Properties

For both growth and decay parent functions $f(x) = b^x$:

- **Domain:** ____________
- **Range:** ____________
- **Horizontal Asymptote:** ____________
- **y-intercept:** ____________
- **Key points on the graph:** $(-1,$ ________$)$, $(0,$ ________$)$, and $(1,$ ________$)$.

Transformation Form

The standard transformation form of an exponential function is

$$f(x) = a(b)^{x-h} + k$$

where:

- h controls the ________ ________ (shifts right if $h > 0$, left if $h < 0$).
- k controls the ________ ________ and shifts the horizontal asymptote to ________.
- a controls reflection and scaling:
  - If $a < 0$, the graph is ________ over the x-axis.
  - If $|a| > 1$, the graph is vertically ________.
  - If $0 < |a| < 1$, the graph is vertically ________.

Example 1

Analyze the graph of $f(x) = \left(\frac{1}{2}\right)^x - 3$, then sketch it.

Example 2

Analyze the graph of $f(x) = 4^{x-1} + 2$, then sketch it.

Example 3

Find the exponential function $f(x) = ab^x$ such that $f(1) = -6$ and $f(2) = -18$.

Example 4

Find the exponential function $f(x) = ab^{-x} + c$ that has a horizontal asymptote at $y = 32$ and passes through the points (0, 122) and (2, 72).

The Natural Base e

The **natural base** ________ is an irrational number defined as the limit of $\left(1 + \frac{1}{x}\right)^x$ as $x \to \infty$:

$$e \approx 2.718281828459...$$

The function $f(x) = e^x$ is called the **natural exponential function**.

Example 5

Let $f(x) = e^{3x} - 2$ and $g(x) = 2x^3 + 3$. Find the value of $(g \circ f)(0)$.

Example 6

The US population (in millions) can be modeled by $P(t) = 331.5e^{0.0079t}$, where t represents the number of years since 2020. Estimate the US population in the year 2040.

Fact

Compound Interest Formulas

Let P be the principal, r be the annual interest rate (as a decimal), t be the time in years, and A be the balance.

- **Compounded n times per year:**

$$A = P\left(1 + \frac{r}{n}\right)^{nt}$$

- **Compounded continuously:**

$$A = Pe^{rt}$$

Example 7

An amount of \$12,000 is invested at a 3% annual interest rate for 4 years. Find the balance in the account if the interest is compounded:
(alph*) quarterly
(alph*) continuously

Homework: Section 3.1 Homework Exercises

3.2 Logarithmic Functions and Their Graphs

Objectives

- Convert between exponential and logarithmic forms.
- Simplify logarithmic and exponential expressions using basic properties.
- Find the inverse of a logarithmic equation.
- Identify key features of logarithmic function graphs (VA, x-intercept, domain, and range).
- Sketch transformed logarithmic functions using a 7-step procedure.

Logarithmic Functions

For $x > 0$, $b > 0$, and $b \neq 1$, the **logarithmic function** with base b is defined as:

$$y = \log_b x \quad \text{if and only if} \quad b^y = x$$

The value y is the exponent to which the base b must be raised to obtain x. In other words, a logarithm is an ________.

Common and Natural Logarithms

- **Common Logarithm:** A logarithm with base ________. It is typically written without the base:

$$\log x = \log_{10} x$$

- **Natural Logarithm:** A logarithm with base ________. It is written as:

$$\ln x = \log_e x$$

Example 1

Convert between forms:
(alph*) Convert $\log_m p = q$ to exponential form.
(alph*) Convert $z^y = w$ to logarithmic form.

Fact

Basic Properties of Logarithms

Let $b > 0$, $b \neq 1$, and x be any real number:

Property	**General Form**	**Natural Log Form**
1. Log of 1 is 0	$\log_b 1 =$ ________	$\ln 1 =$ ________
2. Log of base is 1	$\log_b b =$ ________	$\ln e =$ ________
3. Inverse property	$\log_b b^x =$ ________	$\ln e^x =$ ________
4. Inverse property	$b^{\log_b x} =$ ________	$e^{\ln x} =$ ________
5. One-to-One property	If $\log_b x = \log_b y$, then ________	If $\ln x = \ln y$, then ________

Example 2

Simplify or solve:
(alph*) Evaluate $\log_2 2^5$.
(alph*) Evaluate $7^{\log_7 14}$.
(alph*) Evaluate $\log 10$.
(alph*) Solve $\log_2 x = \log_2 3$ for x.

Example 3

Evaluate or simplify the natural log expressions:

(alph*) $\ln\left(\frac{1}{e}\right)$

(alph*) $e^{\ln 5}$

(alph*) $4\ln 1$

(alph*) $2\ln e$

Fact

Inverse Relationship

Because the exponential function $f(x) = b^x$ and the logarithmic function $g(x) = \log_b x$ are ________ of each other:

- Their graphs are reflections of each other across the line ________.
- The domain of $f(x)$ is the range of $g(x)$, and vice versa.
- To find the algebraic inverse of a function, swap ________ and ________ and solve for the new y.

Example 4

Find the inverse of the function $g(x) = -2 + \log_3(x+1)$.

Transformation of Logarithmic Functions

The standard transformation form of a logarithmic function is

$$f(x) = a\log_b(x-h) + k$$

where:

- h shifts the graph horizontally and moves the vertical asymptote to ________.
- k shifts the graph vertically.
- $a < 0$ reflects the graph over the ________.
- $|a| > 1$ stretches the graph vertically, and $0 < |a| < 1$ compresses it.

Note

7-Step Graphing Procedure

To graph a transformed logarithmic function $f(x) = a\log_b(x-h) + k$:

1. Identify and graph the vertical asymptote: ________.
2. Identify the transformations.
3. Write the equation of the parent graph's inverse: ________.
4. Make a t-table for the inverse using $x = -1, 0, 1$.
5. Reverse the t-table coordinates $(x, y) \to (y, x)$ to get key points for the parent log function $\log_b x$.
6. Adapt the t-table coordinates according to the transformations.
7. Plot the points and sketch the curve.

Example 5

Analyze and sketch the graph of $g(x) = \log_6(x-1) + 4$.

Example 6

Analyze and sketch the graph of $g(x) = -\log_3(x+1) - 3$.

Homework: Section 3.2 Homework Exercises

3.3 Properties of Logarithms

Objectives

- Use the Product, Quotient, and Power properties to evaluate or rewrite logarithmic expressions.
- Expand and condense logarithmic expressions using properties.
- Use the Change of Base property to rewrite and evaluate logarithms with different bases.

Properties of Logarithms

Let b, u, and v be positive real numbers such that $b \neq 1$, and let k be any real number.

- **Product Property:**

$$\log_b(uv) = \underline{\hspace{4cm}}$$

- **Quotient Property:**

$$\log_b\left(\frac{u}{v}\right) = \underline{\hspace{4cm}}$$

- **Power Property:**

$$\log_b(u^k) = \underline{\hspace{4cm}}$$

Note

Common Errors in Logarithm Properties

Be careful to avoid these extremely common algebraic mistakes:

- **Incorrect Product Splitting:** $\log_b(u \cdot v) \neq \log_b u \times \log_b v$
- **Incorrect Quotient Splitting:** $\log_b\left(\frac{u}{v}\right) \neq \frac{\log_b u}{\log_b v}$
- **Incorrect Sum Splitting:** $\log_b(u + v) \neq \log_b u + \log_b v$
- **Incorrect Difference Splitting:** $\log_b(u - v) \neq \log_b u - \log_b v$

Example 1

Simplify the expression using the properties: $\log_6 2 + \log_6 3$

Example 2

Simplify the expression using the properties: $\log_4 32 - \log_4 2$

Example 3

Simplify the expression using the properties: $\log_5 \sqrt{5}$

Condensing Logarithmic Expressions

To **condense** a logarithmic expression means to write a sum or difference of multiple logarithms (with matching bases) as a single logarithm.

*Note: Apply the Power Property to move coefficients to exponents **before** using the Product or Quotient properties.*

Example 4

Condense the expression: $\log_7 a + 2\log_7 d - \log_7 3$

Example 5

Condense the expression: $2\log x - \log w - \log z$

Example 6

Condense the expression: $\frac{1}{3}\left[\log_2(x-4) - 4\log_2 x - \log_2 y\right]$

Expanding Logarithmic Expressions

To **expand** a logarithmic expression means to write a single logarithm containing products, quotients, or powers as a sum, difference, or multiple of simpler logarithmic terms.

Example 7

Expand the expression: $\ln\left(3x^2y^5\right)$

Example 8

Expand the expression: $\log\left(\dfrac{x^3y^2}{\sqrt{z+1}}\right)$

Example 9

Write $\ln 6$ in terms of $\ln 2$ and $\ln 3$.

Example 10

Write $\ln\left(\dfrac{2}{27}\right)$ in terms of $\ln 2$ and $\ln 3$.

Fact

Change of Base Formula

For any positive real numbers a, b, and x where $a \neq 1$ and $b \neq 1$:

$$\log_b x = \frac{\log_a x}{\log_a b}$$

Typically, we convert to base 10 (common log) or base e (natural log):

$$\log_b x = \underline{\hspace{5cm}} = \underline{\hspace{5cm}}$$

Example 11

Use the Change of Base formula with common logarithms to evaluate: $\log_7 8$ (round to the nearest thousandth).

Example 12

Use the Change of Base formula with natural logarithms to evaluate: $\log_2 12$ (round to the nearest thousandth).

Homework: Section 3.3 Homework Exercises

3.4 Solving Exponential and Logarithmic Equations

Objectives

- Solve exponential equations using logarithms.
- Solve logarithmic equations using properties of logarithms.
- Check solutions for extraneous roots.

Solving Exponential Equations

To solve an **exponential equation**:
1. ________ the exponential expression on one side of the equation.
2. Take the ________ (or common logarithm) of both sides.
3. Use the ________ ________ to move the variable exponent out of the exponent position (as a coefficient).
4. Solve for the variable.

Note: Round all final decimal answers to the nearest thousandth.

Example 1

Solve the equation: $4e^{2x} - 3 = 2$

Example 2

Solve the equation: $2\left(3^{2t-5}\right) - 4 = 11$

Example 3

Solve the equation: $5^{2x+1} = 6^{x-2}$

Example 4

Solve the equation: $4\left(2^{x}\right) + 8\left(2^{-x}\right) = 33$

Example 5

Solve the equation: $\dfrac{5^{x} - 5^{-x}}{2} = 3$

Solving Logarithmic Equations

To solve a **logarithmic equation**:
1. ________ all logarithmic terms into a single logarithm on one side of the equation.
2. If the equation has the form $\log_b u = c$, convert it to the ________ ________:

$$b^c = u$$

3. If the equation has the form $\log_b u = \log_b v$, use the ______________ to set $u = v$.
4. Solve for the variable.
5. **Crucial Step:** Always check for ________ ________.

Note

Extraneous Solutions

Because the domain of $f(x) = \log_b x$ is restricted to positive real numbers ($x > 0$), the argument of any logarithm in the original equation must be ________.

If a solved value of x makes the argument of any logarithm ≤ 0, that value is ________ and must be discarded.

Example 6

Solve the equation: $5 + 2\ln x = 4$

Example 7

Solve the equation: $\log_5(2x+3) = \log_5 11 + \log_5 5$

Example 8

Solve the equation: $\log_2 x + \log_2(x+2) = 3$

Example 9

Solve the equation: $\log \sqrt[3]{x} = \sqrt{\log x}$

Homework: Section 3.4 Homework Exercises

3.5 Exponential Applications

Objectives

- Use exponential functions to model and solve real-life problems.

- Model and solve growth and decay, compound interest, doubling time, and half-life problems.

Exponential Growth and Decay Models

A quantity that grows or decays at a rate proportional to its current size can be modeled by:

$$y = ae^{kt}$$

where:
- a is the ________ ________ (when $t = 0$).
- k is the ________ ________ (or continuous growth/decay rate).
  - If $k > 0$, the model represents ________.
  - If $k < 0$, the model represents ________.
- t is time.

Fact

Summary of Applications Formulas

Application	Formula	Key Variable Meanings
Discrete Compound Interest	$A = P\left(1 + \frac{r}{n}\right)^{nt}$	$n =$ compoundings per year, $r =$ decimal interest rate
Continuous Compound Interest	$A = Pe^{rt}$	$r =$ continuous rate
Population Growth or Decay	$y = ae^{kt}$	$k =$ growth rate ($k > 0$) or decay rate ($k < 0$)

Example 1

A deposit of \$500 is made in an account that pays 6.75% interest compounded continuously. How long will it take for the balance to double? (Round to the nearest hundredth of a year).

Example 2

The average cost of tuition at a private college can be modeled by $M(t) = 40,000(1.072)^t$, where t represents the number of years since 2020. According to this model, when will the average tuition cost be double what it was in 2020? (Round to the nearest hundredth).

Example 3

The world population (in millions) can be modeled by $P = 7021.7e^{0.01076t}$, where $t = 0$ represents the year 2010. According to this model, when will the world population reach 10 billion (10,000 million)? (Round to the nearest tenth).

Example 4

A culture starts with 500 bacteria and grows at an hourly rate of 40%.
(alph*) Write an exponential growth model $y = ae^{kt}$ for the bacteria count.
(alph*) Estimate the bacteria count after 10 hours.
(alph*) When will the bacteria count reach 80,000? (Round to the nearest hundredth).

Example 5

Polonium-210 has a half-life of 140 days. An initial sample contains 300 mg of Polonium-210.
(alph*) Find the exponential decay model $y = ae^{kt}$ for the mass of the sample remaining.
(alph*) How long will it take for the sample to decay to 200 mg? (Round to the nearest tenth of a day).

Homework: Section 3.5 Homework Exercises